Topic: Inverse Laplace transforms

Question: Find the inverse Laplace transform.

$$F(s) = \frac{s+5}{s^2+5s+6}$$

Answer choices:

A $f(t) = 3e^{2t} + 2e^{3t}$

B $f(t) = 3e^{-2t} + 2e^{-3t}$

C $f(t) = 3e^{2t} - 2e^{3t}$

D $f(t) = 3e^{-2t} - 2e^{-3t}$

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Solution: D

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s+5}{s^2+5s+6}$$

$$F(s) = \frac{s+5}{(s+2)(s+3)}$$

and then apply a partial fractions decomposition.

$$\frac{s+5}{(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3}$$

$$s+5 = A(s+3) + B(s+2)$$

To find A, we'll set $s = -2$ in order to eliminate B.

$$-2+5 = A((-2)+3) + B((-2)+2)$$

$$3 = A(1) + B(0)$$

$$A = 3$$

To find B, we'll set $s = -3$ in order to eliminate A.

$$-3+5 = A((-3)+3) + B((-3)+2)$$

$$2 = A(0) + B(-1)$$

$$2 = -B$$

$$B = -2$$

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Substituting these values back into the decomposition gives

$$F(s) = \frac{3}{s+2} + \frac{-2}{s+3}$$

$$F(s) = 3\left(\frac{1}{s+2}\right) - 2\left(\frac{1}{s+3}\right)$$

$$F(s) = 3\left(\frac{1}{s-(-2)}\right) - 2\left(\frac{1}{s-(-3)}\right)$$

In this form, we can see that both values inside the parentheses resemble the Laplace transform

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

We can see that $a_1 = -2$ and $a_2 = -3$. We'll reverse the formula from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = 3e^{-2t} - 2e^{-3t}$$

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Topic: Inverse Laplace transforms

Question: Find the inverse Laplace transform.

$$F(s) = \frac{s-2}{s^2+4s}$$

Answer choices:

A $f(t) = -\frac{1}{2}t + \frac{3}{2}e^{-4t}$

B $f(t) = -\frac{1}{2}t + \frac{3}{2}e^{4t}$

C $f(t) = -\frac{1}{2} + \frac{3}{2}e^{-4t}$

D $f(t) = -\frac{1}{2} + \frac{3}{2}e^{4t}$

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Solution: C

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{s-2}{s^2+4s}$$

$$F(s) = \frac{s-2}{s(s+4)}$$

and then apply a partial fractions decomposition.

$$\frac{s-2}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

$$s-2 = A(s+4) + Bs$$

To find A, we'll set $s = 0$ in order to eliminate B.

$$0-2 = A(0+4) + B(0)$$

$$-2 = A(4) + B(0)$$

$$A = -\frac{1}{2}$$

To find B, we'll set $s = -4$ in order to eliminate A.

$$-4-2 = A(-4+4) + B(-4)$$

$$-6 = A(0) + B(-4)$$

$$-6 = -4B$$

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$$B = \frac{3}{2}$$

Substituting these values back into the decomposition gives

$$F(s) = \frac{-\frac{1}{2}}{s} + \frac{\frac{3}{2}}{s+4}$$

$$F(s) = -\frac{1}{2}\left(\frac{1}{s}\right) + \frac{3}{2}\left(\frac{1}{s+4}\right)$$

$$F(s) = -\frac{1}{2}\left(\frac{1}{s}\right) + \frac{3}{2}\left(\frac{1}{s-(-4)}\right)$$

In this form, we can see that the values inside the parentheses resemble the Laplace transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(e^{at}) = \frac{1}{s-a}$$

With $a = -4$, we'll reverse the formulas from the table to rewrite the transform as

$$f(t) = \mathscr{L}^{-1}(F(s)) = -\frac{1}{2} + \frac{3}{2}e^{-4t}$$

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Topic: Inverse Laplace transforms

Question: Find the inverse Laplace transform.

$$F(s) = \frac{3s + 1}{2s^2 + s - 1}$$

Answer choices:

A $f(t) = \frac{5}{3}e^{-\frac{1}{2}t} + \frac{2}{3}e^{-t}$

B $f(t) = \frac{5}{6}e^{\frac{1}{2}t} + \frac{2}{3}e^{-t}$

C $f(t) = \frac{5}{3}e^{-t} + \frac{2}{3}e^{-t}$

D $f(t) = \frac{5}{6}e^{-t} + \frac{2}{3}e^{-t}$

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Solution: B

We'll first factor the denominator of $F(s)$,

$$F(s) = \frac{3s + 1}{2s^2 + s - 1}$$

$$F(s) = \frac{3s + 1}{(2s - 1)(s + 1)}$$

and then apply a partial fractions decomposition.

$$\frac{3s + 1}{(2s - 1)(s + 1)} = \frac{A}{2s - 1} + \frac{B}{s + 1}$$

$$3s + 1 = A(s + 1) + B(2s - 1)$$

To find A, we'll set $s = 1/2$ in order to eliminate B.

$$3\left(\frac{1}{2}\right) + 1 = A\left(\frac{1}{2} + 1\right) + B\left(2\left(\frac{1}{2}\right) - 1\right)$$

$$\frac{5}{2} = A\left(\frac{3}{2}\right) + B(0)$$

$$A = \frac{5}{3}$$

To find B, we'll set $s = -1$ in order to eliminate A.

$$3(-1) + 1 = A((-1) + 1) + B(2(-1) - 1)$$

$$-2 = A(0) + B(-3)$$

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$$-2 = -3B$$

$$B = \frac{2}{3}$$

Substituting these values back into the decomposition gives

$$F(s) = \frac{\frac{5}{3}}{2s-1} + \frac{\frac{2}{3}}{s+1}$$

$$F(s) = \frac{5}{3}\left(\frac{1}{2s-1}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

$$F(s) = \frac{5}{3}\left(\frac{\frac{1}{2}}{\frac{2s}{2}-\frac{1}{2}}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

$$F(s) = \frac{5}{6}\left(\frac{1}{s-\frac{1}{2}}\right) + \frac{2}{3}\left(\frac{1}{s+1}\right)$$

$$F(s) = \frac{5}{6}\left(\frac{1}{s-\frac{1}{2}}\right) + \frac{2}{3}\left(\frac{1}{s-(-1)}\right)$$

In this form, we can see that both values inside the parentheses resemble the Laplace transform

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

We can see that $a_1 = 1/2$ and $a_2 = -1$. We'll reverse the formula from the table to rewrite the transform as

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$$f(t) = \mathscr{L}^{-1}(F(s)) = \frac{5}{6}e^{\frac{1}{2}t} + \frac{2}{3}e^{-t}$$

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